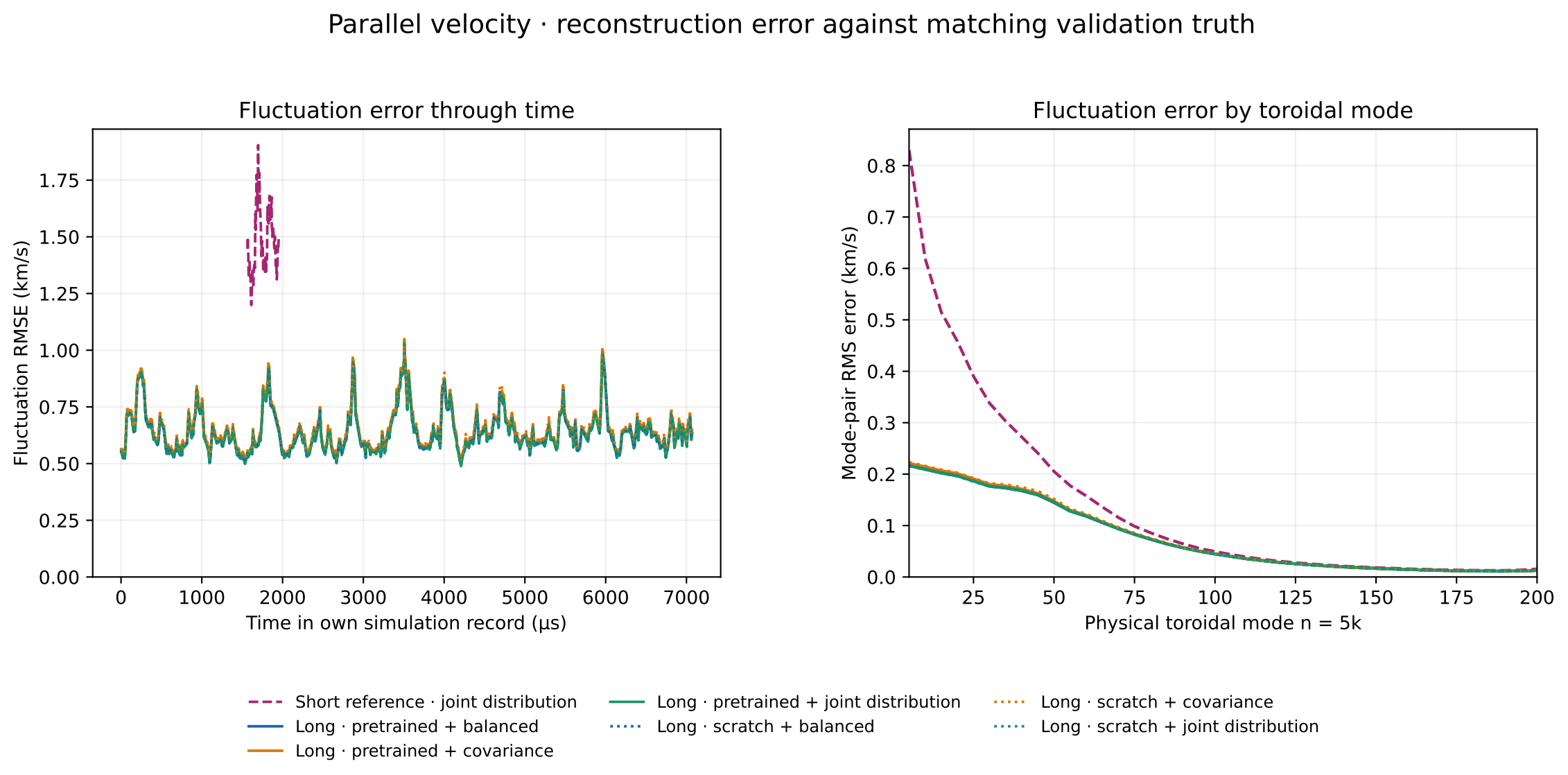
Parallel velocity · reconstruction error against matching validation truth
Fluctuation error through time
Fluctuation RMSE (km/s)
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
0
1000
2000
3000
4000
5000
6000
7000
Time in own simulation record (μs)
Fluctuation error by toroidal mode
Mode-pair RMS error (km/s)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
25
50
75
100
125
150
175
200
Physical toroidal mode n = 5k
Short reference · joint distribution
Long · pretrained + balanced
Long · pretrained + covariance
Long · pretrained + joint distribution
Long · scratch + balanced
Long · scratch + covariance
Long · scratch + joint distribution